STPM Physics · Semester 1 · Chapter 12

TRANSFER OF THERMAL ENERGY

12.1 Conduction, Convection and Radiation

DEFINITIONS & LAWS

- **Conduction** is the transfer of heat through a material without bulk movement of the material, by lattice vibrations of atoms and, in metals, by the collisions of free electrons; it needs a medium.
- **Convection** is the transfer of heat by the bulk movement of a fluid (liquid or gas) caused by density differences; **radiation** is the transfer of heat by electromagnetic waves (mainly infrared) and does not need a medium.
- **Thermal conductivity** k is the rate of heat flow through a unit area of a material per unit temperature gradient, i.e. the heat transferred per unit time across unit area when the temperature gradient is 1 K m^{-1}.
- **Stefan–Boltzmann law**: the total power radiated per unit surface area of a black body is directly proportional to the fourth power of its thermodynamic temperature.
- A **black body** is a perfect absorber and a perfect emitter of radiation at all wavelengths (emissivity e = 1).
- **Steady state**: the temperature at each point does not change with time, so the rate of heat flow is the same through every cross-section in series.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Rate of heat conduction	$\frac{Q}{t} = \frac{kA(\theta_1 - \theta_2)}{L} = -kA\frac{d\theta}{dx}$	$\frac{Q}{t}$ = rate of heat flow (W); k = thermal conductivity (W m^{-1} K^{-1}); A = cross-sectional area (m^2); L = length or thickness (m); θ_1, θ_2 = temperatures of the two faces; $\frac{d\theta}{dx}$ = temperature gradient (K m^{-1})
Thermal resistance	$R = \frac{L}{kA}$	R = thermal resistance (K W^{-1}); L = length or thickness (m); k = thermal conductivity (W m^{-1} K^{-1}); A = cross-sectional area (m^2)
Layers in series	$R = R_1 + R_2 + \ldots$	R = total thermal resistance (K W^{-1}); R_1, R_2, … = thermal resistances of the layers
Heat flow through thermal resistance	$\frac{Q}{t} = \frac{\Delta\theta}{R}$	Δθ = temperature difference (K); R = thermal resistance (K W^{-1})
Power radiated	$P = e\sigma AT^4$	P = power radiated (W); σ = 5.67×10^{-8} W m^{-2} K^{-4}; e = emissivity; A = surface area (m^2); T = temperature (K, in kelvin)

Quantity / law	Formula	Symbols, units & conditions
Net power radiated	$P_{net} = e\sigma A(T^4 - T_0^4)$	body at temperature T in surroundings at temperature T_0 (K); P_{net} = net power radiated (W); e = emissivity; A = surface area (m^2); σ = Stefan–Boltzmann constant
Surface area of a sphere	$A = 4\pi r^2$	for a spherical star or planet; r = radius (m); A = surface area (m^2)
Intensity at a distance	$I = \frac{P}{4\pi d^2}$	I = intensity at distance d (W m^{-2}); P = power emitted (W); d = distance from the source (m)

- Energy balance: at equilibrium, power absorbed = power emitted.

TIPS TO ANSWER

- Conduction: state $\frac{Q}{t} = \frac{kA\Delta\theta}{L}$, then substitute with L in metres and Δθ in K or °C (a difference only), and give the answer in W.
- Composite bar or wall in series: state that the rate of heat flow is the same through each layer in steady state, and write the interface temperature from $\frac{Q}{t}$ equal for the two sections.
- Radiation: convert T to K before using T^4, include e (equal to 1 for a black body), and use the $T^4 - T_0^4$ form for net power.
- For a qualitative explanation, link each mode to its mechanism: free electrons (metals, good conductors), density difference (convection currents), and electromagnetic waves (no medium; the Sun heats the Earth through vacuum).

SECTION A TRAPS

Trap: Using °C in $P = e\sigma AT^4$; the temperature must be in kelvin, but in Δθ for conduction the temperature difference is the same in °C and K.

Trap: Stating that the heat flow differs between layers of different materials in series at steady state; the rate of flow is the same, and it is the temperature gradient that is larger in the poorer conductor.

Trap: Forgetting the fourth power: doubling the thermodynamic temperature increases the power by a factor of 16, not 2 or 4; and using the radius in place of the area $A = 4\pi r^2$ for a sphere.

TRICKS

Trick: Ratios: for the same bar, $\frac{Q}{t} \propto A$ and $\propto \frac{1}{L}$; doubling the radius of a rod gives a 4 times larger heat flow, and doubling the length halves it.

> **Trick:** Radiation ratio: $P \propto AT^4$, so compare two bodies with $\frac{A_1}{A_2}(\frac{T_1}{T_2})^4$ and cancel σ and e when they are the same.